Module 2: Alfvén-Wave Verification and Magnetic-Pressure Pulse

MHD_PINN_proj1 Physics Notes

Revised to be explicit, step-by-step, and derivation-heavy

These notes explain the two benchmark problems used after the baseline ideal-MHD equations are in place: (1) a small-amplitude Alfvén wave for *verification*, and (2) a localized magnetic-pressure pulse for a *nonlinear two-dimensional demonstration*. The emphasis is not just on what the initial conditions are, but on why those problems were chosen, how their physics follows from the ideal-MHD equations, and what diagnostics genuinely test the solver.

1 Symbol and variable table

Symbol	Name	Meaning
ρ_0	background mass density	Uniform density of the unperturbed equilibrium.
p_0	background gas pressure	Uniform thermal pressure of the unperturbed equilibrium.
B_0	background magnetic-field magnitude	Constant background field used in the Alfvén benchmark.
A	perturbation amplitude	Size of the sinusoidal or Gaussian perturbation imposed initially.
k	wave number	Spatial frequency of the sinusoidal Alfvén mode.
λ	wavelength	Spatial period of the Alfvén wave, $\lambda = 2\pi/k$.
v_A	Alfvén speed	Characteristic propagation speed of the shear Alfvén mode.
T	wave period	Time required for the wave to travel one wavelength.
u_z	out-of-plane velocity	Transverse velocity component used by the benchmark Alfvén mode.
B_z	out-of-plane magnetic perturbation	Transverse magnetic perturbation coupled to u_z.
x_0, y_0	pulse-center coordinates	Center of the Gaussian magnetic-pressure pulse.
σ	pulse width	Characteristic radius of the Gaussian pulse.
$\Delta x, \Delta y$	cell widths	Uniform grid spacing in the two coordinate directions.
Δt	time-step size	Explicit time step set by the Courant–Friedrichs–Lewy stability restriction.
L^2 error	root-mean-square error norm	Discrete measure of the mismatch between numerical and reference solutions.

Symbol	Name	Meaning
$\nabla \cdot \mathbf{B}$	magnetic divergence	Quantity that must remain small for a physically acceptable MHD computation.

2 Why these two cases are used together

The project needs more than one test because one benchmark cannot answer every numerical question. The two chosen cases serve different roles:

1. The Alfvén wave is a **verification problem**. It has a known propagation speed and a simple analytical solution. That makes it ideal for asking, "Is the solver reproducing the correct MHD wave physics?"

2. The magnetic-pressure pulse is a **nonlinear demonstration problem**. It does not have a simple closed-form exact solution, but it does produce interpretable two-dimensional dynamics driven by magnetic overpressure. That makes it ideal for asking, "Can the solver evolve a genuinely nonlinear 2D MHD configuration without immediately producing nonsense?"

Important distinction. A verification case checks correctness against known mathematics. A demonstration case checks whether the code can produce believable physics in a more complicated setting. The project needs both.

3 The Alfvén wave from the ideal-MHD equations

3.1 Background state

The benchmark begins from a simple uniform equilibrium:

$$\rho = \rho_0, \qquad p = p_0, \qquad \mathbf{u}_0 = (0,0,0), \qquad \mathbf{B}_0 = (B_0, 0, 0). \tag{1}$$

This means:

- the plasma is initially at rest,
- density and thermal pressure are constant everywhere,
- the magnetic field is straight and points along the x-direction.

That straight background field is important because the Alfvén wave propagates *along* magnetic field lines.

3.2 What kind of perturbation is added

A small transverse perturbation is placed in the out-of-plane components:

$$u_z(x,0) = A\sin(kx), \tag{2}$$

$$B_z(x,0) = -\sqrt{\rho_0}\,A\sin(kx). \tag{3}$$

All other perturbations are zero. This is a very deliberate choice. It means:

- the disturbance varies only along the field direction x,
- the velocity perturbation is perpendicular to $\mathbf{B}_0$,
- the magnetic perturbation is also perpendicular to $\mathbf{B}_0$,
- the disturbance is small enough that linearization is justified.

3.3 Why the restoring force is magnetic tension

The momentum equation from Module 1 contains the magnetic-force density in the form

$$\mathbf{f}_B = -\nabla\left(\frac{B^2}{2}\right) + (\mathbf{B}\cdot\nabla)\mathbf{B} \tag{4}$$

in the normalized units used by the project.

For the benchmark equilibrium, the background field is straight and uniform, so the magnetic-pressure gradient of the background is zero. The physically active force is therefore the field-line-tension response to the transverse bend created by the perturbation. The field behaves like a tension-bearing structure: if a field line is bent sideways, the curvature produces a restoring force that tries to straighten it.

3.4 Derivation of the reduced Alfvén system

We now derive the reduced equations rather than simply stating them.

Start from the ideal-MHD momentum equation in nonconservative form:

$$\rho\frac{D\mathbf{u}}{Dt} = -\nabla p - \nabla\left(\frac{B^2}{2}\right) + (\mathbf{B}\cdot\nabla)\mathbf{B}. \tag{5}$$

Also start from the induction equation:

$$\frac{\partial\mathbf{B}}{\partial t} = \nabla\times(\mathbf{u}\times\mathbf{B}). \tag{6}$$

Now impose the Alfvén-wave assumptions:

$$\rho = \rho_0, \qquad p = p_0, \qquad \mathbf{u} = (0, 0, u_z(x,t)), \qquad \mathbf{B} = (B_0, 0, B_z(x,t)). \tag{7}$$

These assumptions say that the disturbance is one-dimensional in x, transverse in z, and incompressible to leading order.

3.4.1 Step 1: the z-component of the momentum equation

The left side of (**??**) becomes

$$\rho_0\frac{Du_z}{Dt}. \tag{8}$$

Because the perturbation depends only on x and t, and because the background flow in x is zero, the material derivative reduces to

$$\frac{Du_z}{Dt} = \frac{\partial u_z}{\partial t}. \tag{9}$$

Now examine the right side. The pressure gradient has no z-component because the perturbation is transverse and pressure remains uniform to leading order. The magnetic-pressure term also has

no z-component in this linearized geometry, because it points along spatial gradients of B^2, which are along x. The term that survives is the magnetic-tension term:

$$\left[(\mathbf{B}\cdot\nabla)\mathbf{B}\right]_z. \tag{10}$$

Since

$$\mathbf{B}\cdot\nabla = B_0\frac{\partial}{\partial x} + B_z\frac{\partial}{\partial z}, \tag{11}$$

and there is no z-dependence, this becomes

$$\left[(\mathbf{B}\cdot\nabla)\mathbf{B}\right]_z = B_0\frac{\partial B_z}{\partial x}. \tag{12}$$

Therefore the z-component of the momentum equation reduces to

$$\rho_0\frac{\partial u_z}{\partial t} = B_0\frac{\partial B_z}{\partial x}. \tag{13}$$

Equivalently,

$$\frac{\partial u_z}{\partial t} - \frac{B_0}{\rho_0}\frac{\partial B_z}{\partial x} = 0. \tag{14}$$

3.4.2 Step 2: the z-component of the induction equation

Now compute $\mathbf{u}\times\mathbf{B}$:

$$\mathbf{u}\times\mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & u_z \\ B_0 & 0 & B_z \end{vmatrix} = (0,\ u_zB_0,\ 0). \tag{15}$$

Take the curl:

$$\nabla\times(\mathbf{u}\times\mathbf{B}) = \nabla\times(0, u_zB_0, 0). \tag{16}$$

Because the quantity depends only on x,

$$\nabla\times(0, u_zB_0, 0) = \left(0,\ 0,\ \frac{\partial}{\partial x}\right). \tag{17}$$

Since B_0 is constant,

$$\frac{\partial B_z}{\partial t} = B_0\frac{\partial u_z}{\partial x}. \tag{18}$$

3.4.3 Step 3: combine the two equations into a wave equation

Take a time derivative of (**??**):

$$\frac{\partial^2 u_z}{\partial t^2} - \frac{B_0}{\rho_0}\frac{\partial}{\partial x}\left(\frac{\partial B_z}{\partial t}\right) = 0. \tag{19}$$

Substitute (**??**):

$$\frac{\partial^2 u_z}{\partial t^2} - \frac{B_0}{\rho_0}\frac{\partial}{\partial x}\left(B_0\frac{\partial u_z}{\partial x}\right) = 0. \tag{20}$$

Because B_0 is constant,

$$\frac{\partial^2 u_z}{\partial t^2} - \frac{B_0^2}{\rho_0}\frac{\partial^2 u_z}{\partial x^2} = 0. \tag{21}$$

This is the ordinary wave equation with speed

$$v_A = \frac{B_0}{\sqrt{\rho_0}}. \tag{22}$$

The same derivation holds for B_z.

3.5 Exact traveling-wave solution

For the initial sinusoidal mode, the solution is a translated profile:

$$u_z(x,t) = A \sin\left[k(x - v_A t)\right], \tag{23}$$

$$B_z(x,t) = -\sqrt{\rho_0}\, A \sin\left[k(x - v_A t)\right]. \tag{24}$$

The wavelength and period are

$$\lambda = \frac{2\pi}{k}, \qquad T = \frac{\lambda}{v_A}. \tag{25}$$

3.6 Why the solver can be judged quantitatively

Because the benchmark has an explicit analytical solution, the solver can be tested with concrete diagnostics.

3.6.1 L^2 error

For a discrete line cut q_i^{num} and exact values q_i^{exact}, the discrete root-mean-square error is

$$\|e\|_{L^2} = \sqrt{\frac{1}{N_x} \sum_{i=1}^{N_x} \left(q_i^{\text{num}} - q_i^{\text{exact}}\right)^2}. \tag{26}$$

A smaller error means the waveform shape and amplitude are being captured more accurately.

3.6.2 Measured phase speed

The code also measures how far the waveform has shifted in time and compares that against the predicted speed v_A. This is important because a numerical method could, in principle, preserve shape reasonably well but move the wave at the wrong speed. The benchmark therefore tests both *shape fidelity* and *characteristic-speed fidelity*.

3.6.3 Why the final time matters

If the benchmark is run for exactly one full wave period, a periodic sinusoid returns to the same phase modulo the domain length. Then a crude shift-based diagnostic can report zero net displacement even though the wave propagated correctly. For that reason, measuring the phase shift at a quarter period or another nondegenerate fraction of the period is often a better diagnostic choice.

4 Why the solver is upgraded from first order to second order

4.1 What first-order diffusion does to a smooth wave

A first-order piecewise-constant finite-volume scheme with a diffusive numerical flux tends to smear smooth structures. For the Alfvén benchmark, that means:

- the sinusoidal crest becomes too rounded,
- the trough becomes too shallow,
- the amplitude decays more than it should,
- the line-cut comparison against the exact solution becomes visibly blurred.

That is not a failure of the underlying physics model; it is a limitation of the numerical approximation.

4.2 Why piecewise-linear reconstruction helps

Instead of treating every cell as having a constant state, the piecewise-linear method (PLM) reconstructs a linear profile within each cell:

$$q(x) \approx q_i + \sigma_i(x - x_i), \tag{27}$$

where q_i is the cell average and σ_i is a slope estimate. The reconstructed left and right interface states are

$$q^L_{i+1/2} = q_i + \frac{1}{2}\sigma_i \Delta x, \qquad q^R_{i+1/2} = q_{i+1} - \frac{1}{2}\sigma_{i+1}\Delta x. \tag{28}$$

This makes the interface data much more faithful to a smooth sinusoidal field.

4.3 Why a limiter is still necessary

If the slopes are taken naively, oscillations can appear near sharp gradients. The minmod limiter suppresses that by selecting the smaller of the forward and backward slopes when they have the same sign, and setting the slope to zero when they do not:

$$\mathrm{minmod}(a, b) = \begin{cases} \mathrm{sign}(a)\min(|a|, |b|), & ab > 0, \\ 0, & ab \le 0. \end{cases} \tag{29}$$

So the method becomes second-order accurate in smooth regions but retreats toward first order in dangerous regions.

4.4 Why Runge–Kutta 2 is paired with PLM

The spatial approximation alone is not the whole story. If the time update remains only first-order accurate, some of the benefit of the improved spatial reconstruction is lost. The two-stage Runge–Kutta method therefore updates the state via

$$\mathbf{U}^{(1)} = \mathbf{U}^n + \Delta t\, \mathcal{L}(\mathbf{U}^n), \tag{30}$$

$$\mathbf{U}^{n+1} = \frac{1}{2}\left[\mathbf{U}^n + \mathbf{U}^{(1)} + \Delta t\, \mathcal{L}(\mathbf{U}^{(1)})\right], \tag{31}$$

where $\mathcal{L}$ is the finite-volume spatial operator. The physical solution is unchanged; what improves is the fidelity with which the numerical method follows the continuous evolution.

5 The magnetic-pressure pulse as a nonlinear demonstration

5.1 Why magnetic pressure can launch motion

From Module 1, the magnetic force can be written as

$$\mathbf{f}_B = -\nabla\left(\frac{B^2}{2}\right) + (\mathbf{B}\cdot\nabla)\mathbf{B}. \tag{32}$$

The first term behaves like a pressure gradient built from magnetic energy density. Therefore, if one region has a larger B^2 than its surroundings, that region is *magnetically overpressured* and pushes outward on the surrounding fluid.

5.2 How the pulse is initialized

The demonstration case introduces a localized out-of-plane magnetic perturbation:

$$B_z(x, y, 0) = A \exp\left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2}\right]. \tag{33}$$

The density, thermal pressure, and in-plane magnetic field are otherwise taken uniform. The Gaussian profile means the field is strongest near (x_0, y_0) and decreases smoothly away from the center.

5.3 Why this creates a magnetic-pressure gradient

Because B_z is largest at the pulse center, the magnetic energy density

$$e_{\text{mag}} = \frac{1}{2}|\mathbf{B}|^2 \tag{34}$$

is also largest there. Therefore the magnetic-pressure contribution

$$p_{\text{mag}} = \frac{1}{2}|\mathbf{B}|^2 \tag{35}$$

has a spatial gradient pointing outward from the pulse center. The force density contribution

$$-\nabla p_{\text{mag}} \tag{36}$$

therefore pushes fluid away from the region of strongest field.

5.4 Why the pulse is nonlinear

Unlike the Alfvén verification problem, the Gaussian magnetic-pressure pulse is not just a translated copy of itself. As it evolves,

- the field redistributes,
- velocity develops in multiple directions,
- density and gas pressure can change,
- nonlinear couplings between field and flow become important.

So this case asks the solver to do more than follow a single linear wave family. It asks the solver to handle a genuinely two-dimensional MHD response.

Interpretation. The pulse is not mainly about having an exact formula. It is about checking whether the code can produce a physically interpretable nonlinear MHD evolution without obvious pathologies such as exploding divergence error, negative pressure, or wildly asymmetric artifacts unrelated to the problem setup.

5.5 Why the diagnostics change for this case

Since there is no simple pointwise exact solution, the diagnostic focus changes:

1. monitor total mass conservation,
2. monitor total-energy behavior,
3. monitor $\max |\nabla \cdot \mathbf{B}|$,
4. inspect maps of density, pressure, velocity magnitude, and magnetic-field magnitude.

These diagnostics do not prove exact correctness the way the Alfvén benchmark does, but they do check whether the nonlinear evolution remains physically credible.

6 Why magnetic divergence must remain part of the workflow

Both benchmark problems must satisfy

$$\nabla \cdot \mathbf{B} = 0. \tag{37}$$

Even if the solver does not yet enforce this constraint exactly with a specialized method such as constrained transport, it must still monitor the quantity. Large divergence error can feed back into the momentum equation and create unphysical forces. That is why the divergence diagnostic is not optional bookkeeping; it is part of the physics validation logic.

7 What Module 2 accomplishes

After Module 1 introduces the governing equations, Module 2 turns them into a testing strategy:

- the Alfvén wave checks whether the solver reproduces a known MHD propagation problem with the correct speed and low numerical error,
- the magnetic-pressure pulse checks whether the same solver can evolve a nonlinear and spatially two-dimensional magnetic overpressure problem in a believable way,
- the second-order upgrade is motivated by the need to capture the smooth benchmark more faithfully while keeping the method robust near stronger gradients.

These ideas form the bridge between the basic equations and the more advanced numerical and PINN modules that follow.